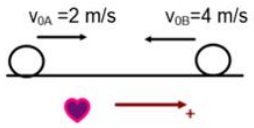
Componente Curricular: **Física – Dinâmica**
Professora: Rosana C. - Pádua **Série:** 2ª **Data:** Outubro
Conteúdo: Choques - Exercícios - Resolução

1-



$Q_{antes} = Q_{depois}$

$m_A.v_{oA} + m_B.v_{oB} = m_A.v_A + m_B.v_B$

$m.2 + m.(-4) = m.v_A + m.v_B$

$-2 = v_A + v_B$

$$e = \frac{v_B - v_A}{v_{0A} - v_{0B}} \Rightarrow 1 = \frac{v_B - v_A}{2-(-4)}$$

$v_B - v_A = 6$

$$\therefore \quad \begin{aligned} -2 &= v_A + v_B \\ 6 &= v_B - v_A \end{aligned}$$

$v_A = -4$ m/s e $v_B = 2$ m/s

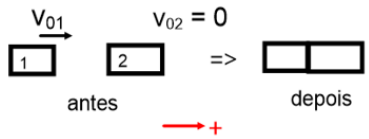
Conclusão



Choques frontais, perfeitamente elásticos entre corpos de massas iguais, as velocidades trocam de valor, em módulo.

2-

a)



$Q_{0sist} = Q_{sist}$

$m_1.v_{01} + m_1.v_{02} = (m_1+m_2).v$

$4.3 = (4+2).v$

$v = 2$ m/s

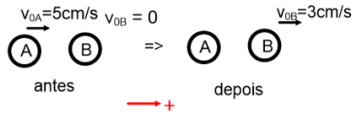
b)

$* E_{cantes} = \frac{m.v_{01}^2}{2} + \frac{m.v_{02}^2}{2}$ => $E_{cantes} = \frac{4.3^2}{2} = 18$ J

$* E_{cdepois} = \frac{m.v_1^2}{2} + \frac{m.v_2^2}{2}$ => $E_{cdepois} = \frac{(4+2).2^2}{2} = 12$ J

$*\Delta Ec = E_{cdepois} - E_{cantes}$ => $\Delta Ec = 12 - 18$ => $\Delta Ec = -6$J

3-



$Q_{0sist} = Q_{sist}$

$m_A.v_{0A} + m_B.v_0 = m_A.v_A + m_B.v_B$

$200.5 = 200.v_A + 300.3$

$v_A = 0,5$ cm/s

$$e = \frac{v_B - v_A}{v_{0A} - v_{0B}} \Rightarrow e = \frac{3 - 0,5}{5 - 0}$$

$$e = \frac{2,5}{5}$$

$$e = 0,5$$

b)

$$* E_{cantes} = \frac{m.v_{0A}^2}{2} + \frac{m.v_{0B}^2}{2} \Rightarrow E_{cantes} = \frac{200.5^2}{2} = 2500 \text{ g.cm}^2/\text{s}^2$$

$$* E_{cdepois} = \frac{m.v_A^2}{2} + \frac{m.v_B^2}{2} \Rightarrow E_{cdepois} = \frac{200.0,5^2}{2} + \frac{300.3^2}{2} = 1375 \text{ g.cm}^2/\text{s}^2$$

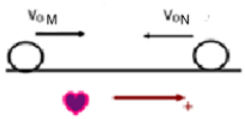
$*E_{dis} = E_{cantes} - E_{cdepois} \Rightarrow E_{dis} = 2500 - 1375 \Rightarrow E_{dis} = 1125 \text{ g.cm}^2/\text{s}^2$

$*2500 \text{ g.cm}^2/\text{s}^2$ ------- 100%

$1125 \text{ g.cm}^2/\text{s}^2$ ------- x

$\Rightarrow$ x = 45%

4-



$Q_{0sist} = Q_{sist}$

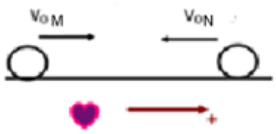
$m_M.v_{0M} + m_N.v_{0N} = m_M.v_M + m_N.v_N$

$15.5 + 13.(-3) = 15.v_M + 13.v_N$

$15.v_M + 13.v_N = 36$

$$e = \frac{v_M - v_N}{v_{0N} - v_{0M}} \Rightarrow \frac{3}{4} = \frac{v_M - v_N}{-3 - (5)}$$

$$v_M - v_N = -6$$



$15.v_M + 13.v_N = 36$

$v_M - v_N = -6$ (x13)

$13v_M - 13v_N = -78$

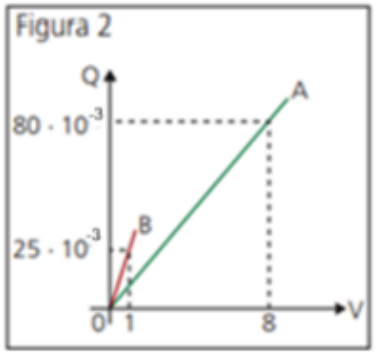
$15.v_M + 13.v_N = 36$

$28.v_M = -42$

$v_M = -1,5$ m/s para a esquerda

$v_N = +4,5$ m/s para a direita

5-



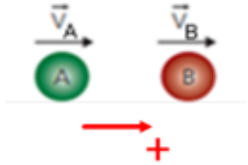
$Q = m.v$

*Bola B: $25.10^{-3} = m_B.1$

$m_B = 25.10^{-3}\ kg$

*Bola A: $80.10^{-3} = m_A.8$

$m_A = 10.10^{-3}\ kg$



$m_A.v_{0A} + m_B.V_{0B} = (m_A + m_B).v$

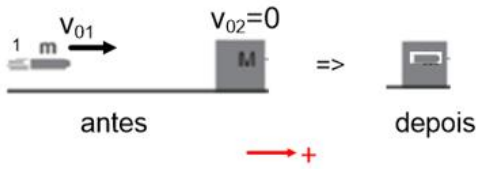
$10.10^{-3}.8 + 25.10^{-3}.1 = (10.10^{-3} + 25.10^{-3}).v$

$105.10^{-3} = 35.10^{-3}.V$

$v = 3\ m/s$

6-

1- Colisão



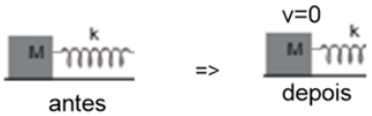
$Q_{antes} = Q_{depois}$

$m_A.v_{oA} + m_B.v_{oB} = (m_A + m_B).v$

$12.200 = 144.v$

$v = \frac{50}{3}\ m/s$

2- Deformação



$E_{Mantes} = E_{Mdepois}$

$E_{c0} = E_{pe}$

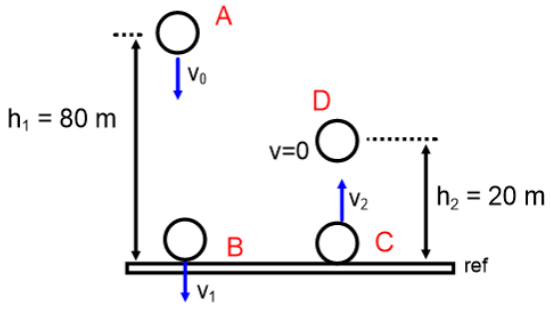
$$\frac{(m_A + m_B).v^2}{2} = \frac{k.x^2}{2}$$

$$0{,}144.\left(\frac{50}{3}\right)^2 = 1{,}6.10^4.x^2$$

$x^2 = 0{,}0025$

$x = 0{,}05\ m \Rightarrow x = 5\ cm$

7-



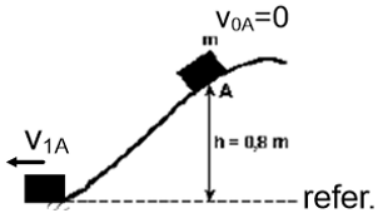
$$e = \sqrt{\frac{h_2}{h_1}} \qquad e = \sqrt{\frac{20}{80}}$$

$$e = \sqrt{\frac{1}{4}} = \sqrt{0{,}25}$$

$$e = 0{,}5$$

8-

1- Queda do bloco



*E_{Mantes} = $E_{Mdepois}$

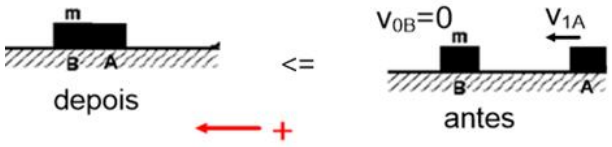
E_{p0} = E_c

$$m.g.h = \frac{m.v_{1A}^2}{2}$$

$$2.10.0{,}8 = v_{1A}^2$$

$$v_{1A}^2 = 16$$

2- Colisão



$$Q_{0sist} = Q_{sist}$$

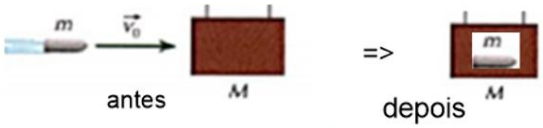
$$m.v_{1A} + m.v_{0B} = 2m.v$$

$$4 = 2.v$$

$$v = 2 \text{ m/s}$$

9-

1- Colisão



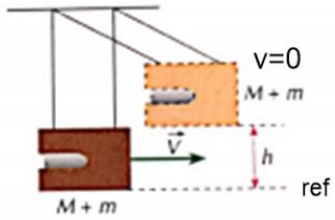
$\rightarrow +$

$Q_{0sist} = Q_{sist}$

$m_1.v_{01} + m_{01}.v_{02} = (m_1+m_2).v$

$20.v_0 = 2020.v \Rightarrow v_0 = 101.v$

2- Subida



$$E_c = E_p$$

$$\frac{(M+m)\cdot V^2}{2} = (M+m)\cdot gh$$

$$V = \sqrt{2gh}$$

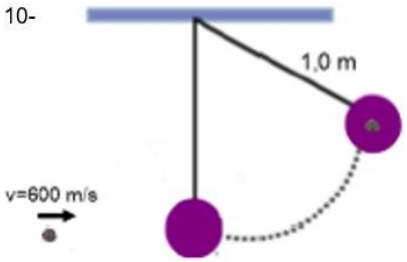
$$V = \sqrt{2.10.0,2}$$

V=2 m/s

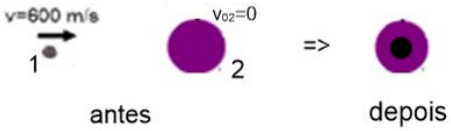
$v_0 = 101.v \Rightarrow v_0 = 101.2$

$v_0 \cong 200$ m/s

10-



1- Colisão m=0,01 kg; M=6 kg



$\rightarrow +$

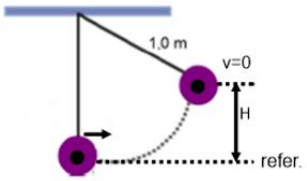
$Q_{0sist} = Q_{sist}$

$$m\cdot v = (M+m)\cdot V$$

$$0,01\cdot 600 = (6+0,01)\cdot V$$

$$\boxed{V \approx 1m/s}$$

2- Subida



$$E_C = E_P$$

$$\frac{m \cdot V^2}{2} = m \cdot g \cdot H$$

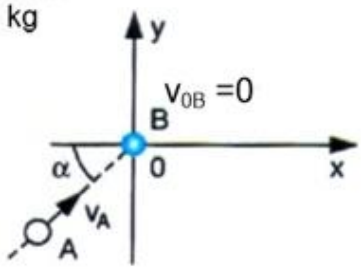
$$H = \frac{V^2}{2 \cdot g}$$

$$H = \frac{1^2}{2 \cdot 10}$$

$$\boxed{H = 0{,}05m}$$

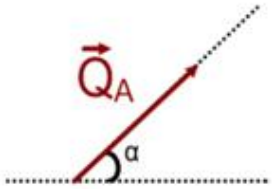
11-

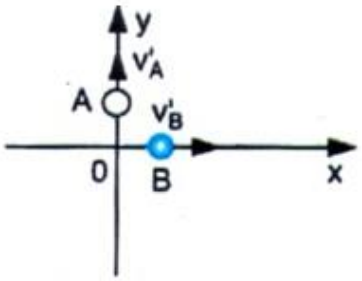
$v_A = 5$ m/s
$m_A = 1$ kg



antes

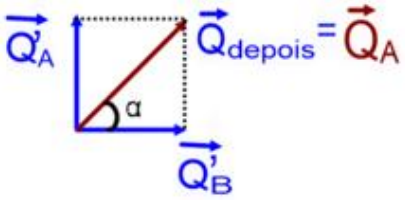
$$\vec{Q}_{antes} = \vec{Q}_A$$





depois

$$\vec{Q}_{antes} = \vec{Q}_{depois}$$



$$\cos\alpha = \frac{Q'_B}{Q_A} \Rightarrow \frac{8}{10} = \frac{m_B.3}{1.5}$$

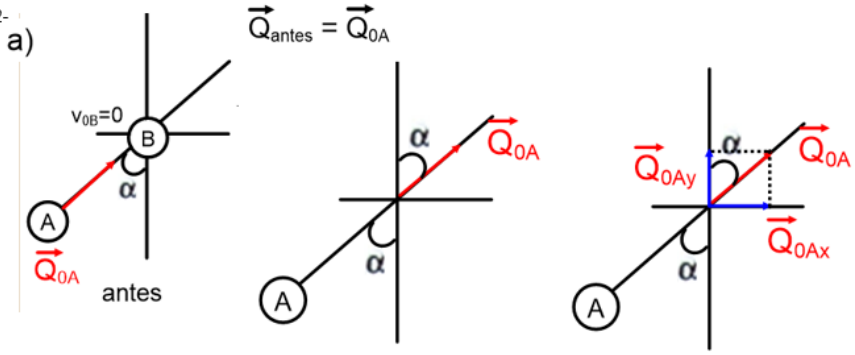
$$m_B = \frac{4}{3}\,kg = 1{,}33\ \text{kg}$$

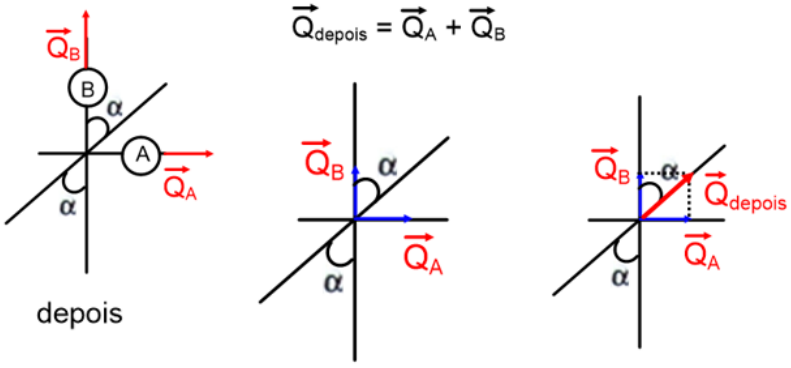
$$sen\,\alpha = \frac{Q'_A}{Q_A} \Rightarrow \frac{6}{10} = \frac{1.v'_A}{1.5}$$

$$v'_A = 3\ \text{m/s}$$

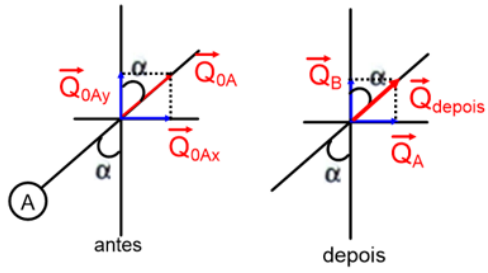
12-

a)





$\vec{Q}_{antes} = \vec{Q}_{depois}$ $(Q_{antes} = Q_{depois})$



*Horizontal: $Q_{0Ax} = Q_A$

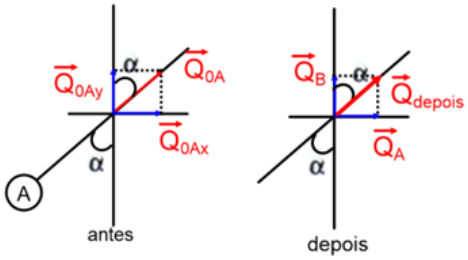
$Q_{OA}.sen\alpha = Q_A$

$m_A.v_{OA}.sen\alpha = m_A.v_A$

$2.0,8 = v_A$

$v_A = 1,6$ m/s

$\therefore v_{Ax} = 1,6$ m/s e $v_{Ay} = 0$



*Vertical: $Q_{0Ay} = Q_B$

$Q_{OA}.cos\alpha = Q_B$

$m_A.v_{OA}.cos\alpha = m_B.v_B$

$2.0,6 = v_B$

$v_B = 1,2$ m/s

$\therefore v_{Bx} = 0$ e $v_{By} = 1,2$ m/s

b) * $E_{cantes} = \frac{m.v_{0A}^2}{2} \Rightarrow E_{cantes} = \frac{0{,}2.2^2}{2} \Rightarrow E_{cantes} = 0{,}4J$

* $Ec_{depois} = \frac{m.v_{Ax}^2}{2} + \frac{m.v_{BY}^2}{2} \Rightarrow E_{cdepois} = \frac{0{,}2.1{,}6^2}{2} + \frac{0{,}2.1{,}2^2}{2}$

* $Ec_{depois} = 0{,}256 + 0{,}144 \Rightarrow Ec_{depois} = 0{,}4J$

* $\Delta E_c = E_{cdepois} - E_{cantes} \Rightarrow * \Delta E_c = 0$